

Element H: Prototype Testing and Data Collection Plan

Testing Details (H1, H2, H3, H4)

	requirement	testing setup and procedure	test cases and number of trials	expected range of results	pass/fail criteria, goal
High Priority 1	Protect the camera	Put the cover on the camera and see if the lens gets damaged with it on	3	Camera is fully protected, camera still has damage on it but less now, camera has same amount of damage on it as before	Lens remains undamaged
2	Stay on the lens when on	Shake the camera when the cover is on, see if the cover falls off or not	3	Cover stays on, cover falls off	No slippage off of the lens
3	Stay attached to the camera	Move the camera around a lot and see if the cover falls off or not	3	Cover stays on, cover falls off	The cover doesn't detach from the camera
4	Easily removable and attachable	Attempt to attach and remove it	5	Cover is easily attachable and removable, cover is attachable and	The cover doesn't stick to the lens for more than 0.5 seconds

				removable with some difficulty, cover can't be attached or removed from the camera	
5	Durability	Attempt to damage it in ways that would occur naturally to a user of the product	5	Cover has no scratches on it, cover is mildly damaged, cover is very fragile	There are no major chips or cracks in the cover
6	Black color	Have the camera record with the cover on and see if the output is dark enough	2	Cover is completely black, cover is moderately dark, cover is not dark at all	The cover is a solid black in color
7	Low weight	Put on scale	2	<50 grams, <100 grams, >100 grams	The lens cover weighs less than 50 grams

Feedback From Field Experts *(H5)*

Revised Testing Details That Incorporate Field Expert Feedback *(H5)*

We showed the design to a field expert, and they really liked the design. We fit the product onto the camera, and it held secure and provided complete coverage of the lens. We even shook it, and the cover held firm. The field expert says that as long as it covers the lens and can stay put, it works. Because of this, it was wholefully approved by the field expert.